



PDF to Audio Converter with Summarizer

Making Reading Accessible and Easier

B.AKHIL AP23110011495

J.PRAMODH AP23110011529

S.PRADEEP AP23110011524

Project Overview

- Python-based application
- Extracts text from PDFs or DOCX
- Converts text to speech audio
- Listen instantly or download MP
- Summarizes PDF content



Motivation



Accessibility

Support visually impaired users



Multitasking

Enable audio for busy users



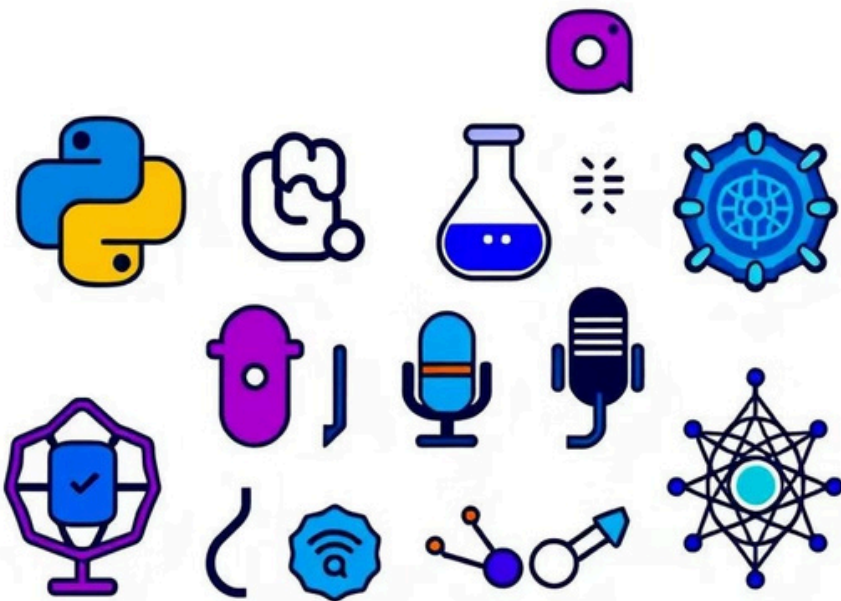
Efficiency

Simplify large document reading



Learning Speed

Faster content consumption



Technologies and Libraries

Backend & Extractors

- Flask web server
- PyMuPDF for PDFs
- python-docx for DOCX

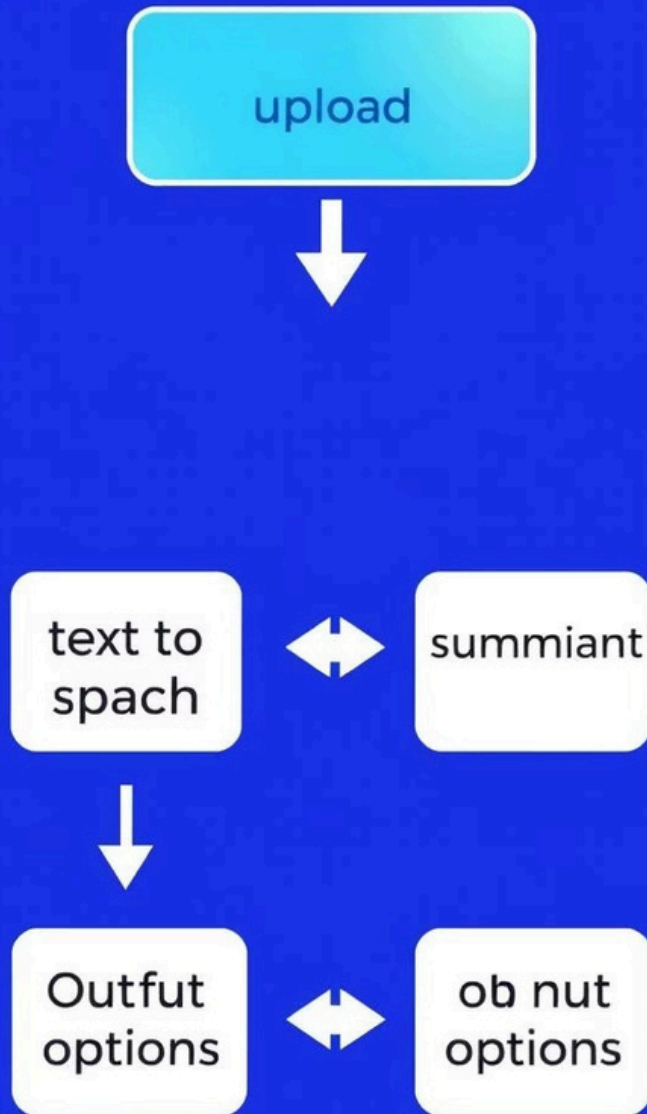
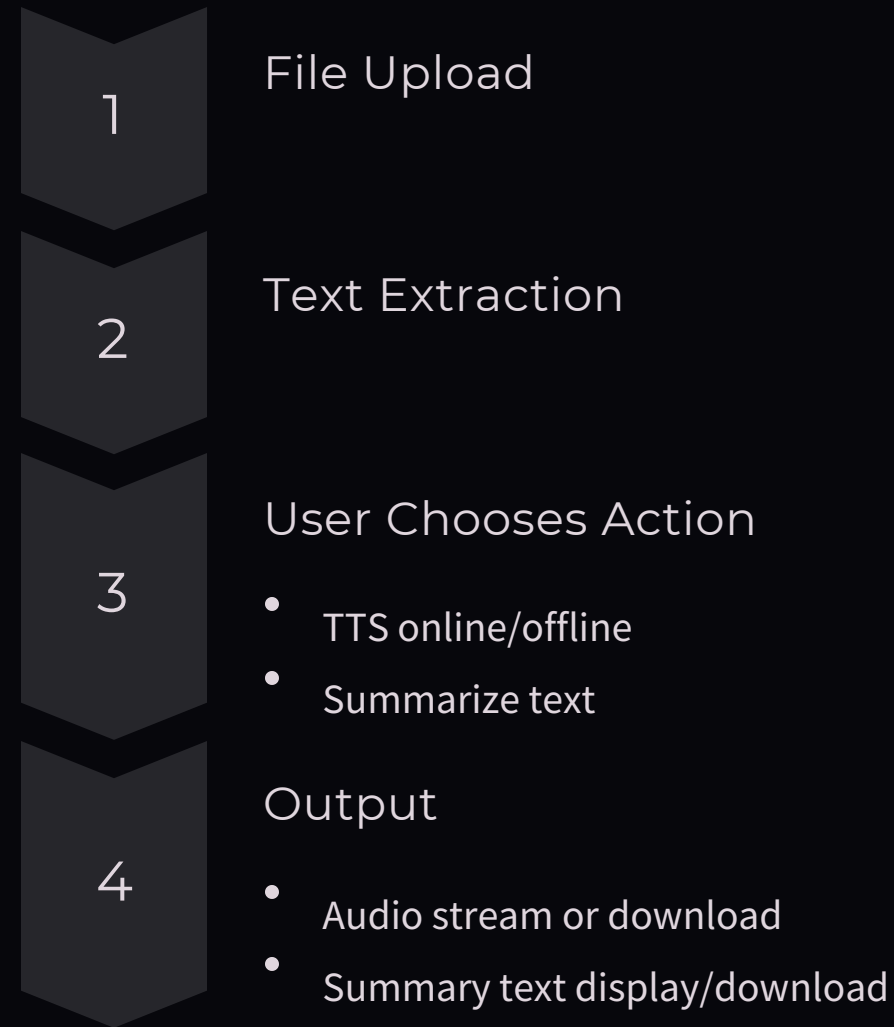
Speech and Summarization

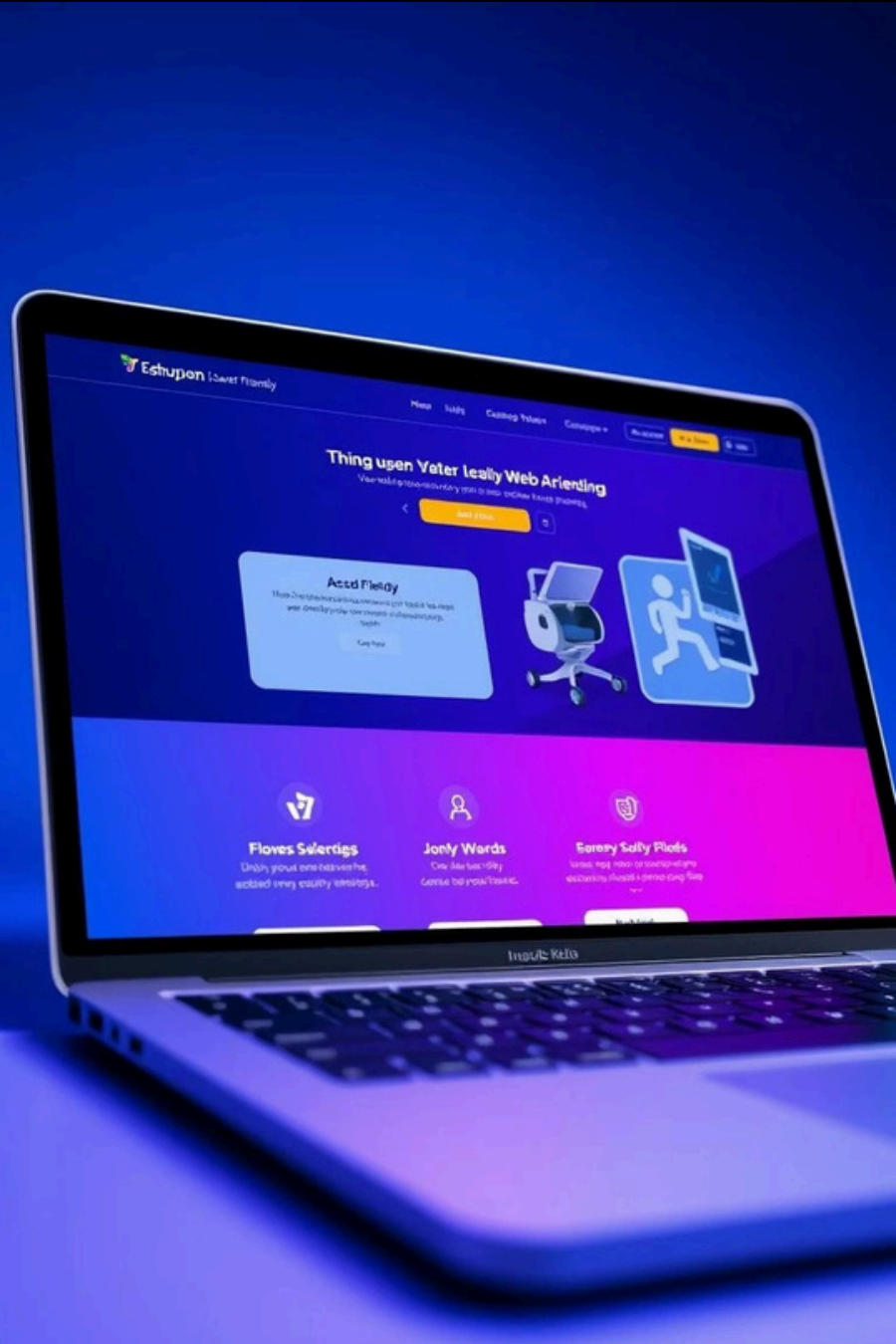
- gTTS online TTS
- pyttsx offline TTS
- Huggingface Transformers

Frontend & Utilities

- HTML / CSS UI
- pydub for audio management

System Architecture





Key Features



PDF & DOCX
Support



Online TTS
(gTTS)



Offline TTS
(pyttsx3)



ML
Summarization

Additional features include downloadable MP , responsive UI, and error handling.

User Interface Overview

- File upload for audio and summary
- TTS mode & language dropdowns
- Submit buttons for conversion
- Theme toggle (Light/Dark/Contrast)

Code Structure

Core Functions

- `extract_text()`
- `convert_text_to_speech()`
- `summarize_text()`

Routes & Templates

- Flask routing
- index.html frontend
- Organized folders (Uploads, Outputs)

[illegible]



Summarizer Module Details



Uses Huggingface
Transformers



Automatic text
chunking for large
files



Outputs saved as .txt files

Example summary extracted from a DBMS PDF shows concise key points.

Challenges & Future Enhancements

Challenges

- Handling large PDFs
- Summarizer limits on short text
- Cross-platform offline TTS
- Improving accessible UI

Future Enhancements

- Add EPUB and TXT support
- Selective page extraction
- Voice customization
- Multi-language summarization
- Support larger files

Continuing to refine accessibility and functionality for all users.